Statistics 5810/6810:
Reproducible Research (RR)

RR Includes

- Complete description of the data and the analysis of the data, including computer programs
- Results need to be exactly reproducible by other researchers (and a researcher could also apply exactly the same analysis steps to a new data set)

RR in Practice

- MD Anderson Cancer Center internally requires RR for their research (see examples at http://bioinformatics.mdanderson.org/Supplements/ReproRsch-All/)

- General discussion at google: https://groups.google.com/forum/?hl=en&fromgroups#!forum/reproducible-research

Tools for RR in R: Sweave (1)

Sweave is a tool that allows to embed the R code for complete data analyses in latex documents. The purpose is to create dynamic reports, which can be updated automatically if data or analysis change. Instead of inserting a prefabricated graph or table into the report, the master document contains the R code necessary to obtain it. When run through R, all data analysis output (tables, graphs, etc.) is created on the fly and inserted into a final latex document. The report can be automatically updated if data or analysis change, which allows for truly reproducible research.

(from: http://www.stat.uni-muenchen.de/~leisch/Sweave/)

Tools for RR in R: Sweave (2)

Additional links for Sweave:

–http://www.stat.uni-muenchen.de/~leisch/Sweave/Sweave-Rnews-2002-3.pdf

–http://www.math.usu.edu/~adele/s6100/sweave.ppt (general & brief)

–http://www.math.ualberta.ca/~mlewis/links/the_joy_of_sweave_v1.pdf (detailed & long)

Sweave itself may be a bit tedious to use in the beginning - but fortunately, we have RStudio!

Tools for RR in R: knitr (1)

knitr: A general-purpose package for dynamic report generation in R

This package provides a general-purpose tool for dynamic report generation in R, which can be used to deal with any type of (plain text) files, including Sweave, HTML, Markdown and reStructuredText. The patterns of code chunks and inline R expressions can be customized. R code is evaluated as if it were copied and pasted in an R terminal thanks to the evaluate package (e.g. we do not need to explicitly print() plots from ggplot2 or lattice). R code can be reformatted by the formatR package so that long lines are automatically wrapped, with indent and spaces being added, and comments being preserved. A simple caching mechanism is provided to cache results from computations for the first time and the computations will be skipped the next time…

(from: http://cran.r-project.org/web/packages/knitr/index.html)

Tools for RR in R: knitr (2)

Additional links for knitr:

–http://yihui.name/knitr/

–http://yihui.name/knitr/demo/minimal/

Again, it is easiest to work from within RStudio!